Practice workbook · Calculate and express a ratio in a required form

Two values, counted, read from a graph or table, are compared as a ratio in simplest form, 1 : n, n : 1, or as how many times greater.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-RATIO

1. Question 5 · part a.i · 2 marks

November 2024 Paper 1

Worked-example retry · Attempt part a.i only.

5 (a) The table gives information about some blood components in three people.

Blood component	Healthy person number per cm^3	Person A number per cm^3	Person B number per cm^3
white blood cells	7500	13 750	7250
red blood cells	5.0×10^6	5.0×10^6	5.0×10^6
platelets	3.0×10^5	3.0×10^5	0.8×10^5

(i) Calculate the ratio of the number of white blood cells in the healthy person compared with the number of white blood cells in person A.
Write your answer in the form of 1 : n

(2)

1 : .....

(ii) Explain a possible reason for the difference in the number of white blood cells in the healthy person and person A.

(2)

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(iii) Describe the effect that the lower number of platelets per cm^3 would have on the health of person B.

(2)

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1. Question 5 · part a.i · 2 marks · Working and answer

Working and answer

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2. Question 7 · part a.i · 2 marks

January 2020 Paper 1

Further practice · Attempt part a.i only.

7 The table gives the percentage of overweight people in different age groups for six countries.

Country	Percentage overweight		Percentage overweight for each age group					
	Males	Females	18–24	25–34	35–44	45–64	65–74	75 years and over
Croatia	67.5	48.3	22.3	38.1	53.7	69.2	73.1	65.7
Malta	66.8	55.2	36.0	52.5	62.7	69.0	74.1	65.3
Poland	64.1	46.7	19.3	38.7	52.8	66.0	73.2	65.5
Portugal	57.6	50.0	23.6	35.1	49.9	63.5	69.3	71.2
Turkey	56.4	56.3	22.4	44.2	64.9	75.4	66.2	56.5
UK	60.2	51.8	29.0	47.3	56.0	64.0	62.8	52.6

(Source from: https://ec.europa.eu/eurostat/statistics-explained/images/5/59/Share_of_overweight_population_by_sex_and_age%2C_2014.png)

(a) (i) The ratio of overweight females to overweight males in the UK is 1:1.2

Calculate the ratio of overweight females to overweight males in Croatia.

Write your answer in the form of 1: n

(2)

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ratio = ...

(ii) Give two trends shown in the data.

(2)

1 .....

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2 .....

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(iii) Give two reasons why the data for a country might be unreliable.

(2)

1 .....

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2 .....

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2. Question 7 · part a.i · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

6 *Staphylococcus aureus* (*S. aureus*) and MRSA are microorganisms that cause a wide range of infections in humans.

The table shows the number of deaths from *S. aureus* and from MRSA in England in one year.

The data is for males and females of different age groups.

Age group in years	Number of deaths in one year per million people			
	Males		Females	
	S. aureus	MRSA	*S. aureus*	MRSA
under 45	1.8	0.5	1.4	0.4
45–54	6.0	2.5	4.3	2.4
55–64	15.7	8.2	8.1	4.2
65–74	41.2	26.1	21.9	12.5
75–84	128.6	86.8	81.1	51.6
85 and over	416.6	299.9	200.3	146.4

Source: *Office for National Statistics*

(a) (i) The data shows that there are more deaths caused by *S. aureus* than by MRSA.

Give two other conclusions that can be made from the data.

(2)

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2 .....

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3. Question 6 · part a.ii · 2 marks

January 2020 Paper 2

Further practice · Attempt part a.ii only.

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(ii) Calculate the ratio of males aged 85 and over who die from *S. aureus* infection to males under the age of 45 who die from *S. aureus* infection.

(2)

ratio = ...

(iii) The total population in England is 53 million.

Calculate the number of people aged between 65 and 74 who died from *S. aureus*.

(2)

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number = ...

(iv) The number of deaths caused by MRSA is lower than the number of deaths caused by *S. aureus*.

Explain why an infection caused by MRSA is still a major concern.

(2)

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3. Question 6 · part a.ii · 2 marks · Working and answer

Working and answer

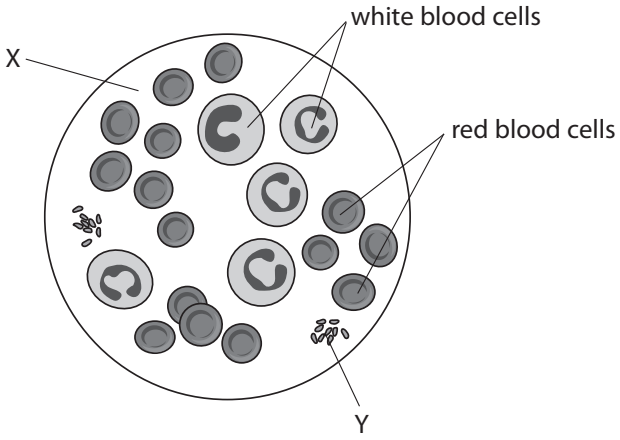
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4. Question 1 · part c.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c.i only.

1 The diagram shows a sample of human blood.



(a) State the function of red blood cells.

(1)

.....

.....

(b) Give the name of X and Y.

(2)

X .....

Y .....

(c) (i) Calculate the ratio of red blood cells to white blood cells.

Write the ratio in its simplest form.

(2)

ratio = ...

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4. Question 1 · part c.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part c.i only.

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(ii) Suggest why this ratio will change if a person has a bacterial infection.

(2)

(Total for Question 1 = 7 marks)

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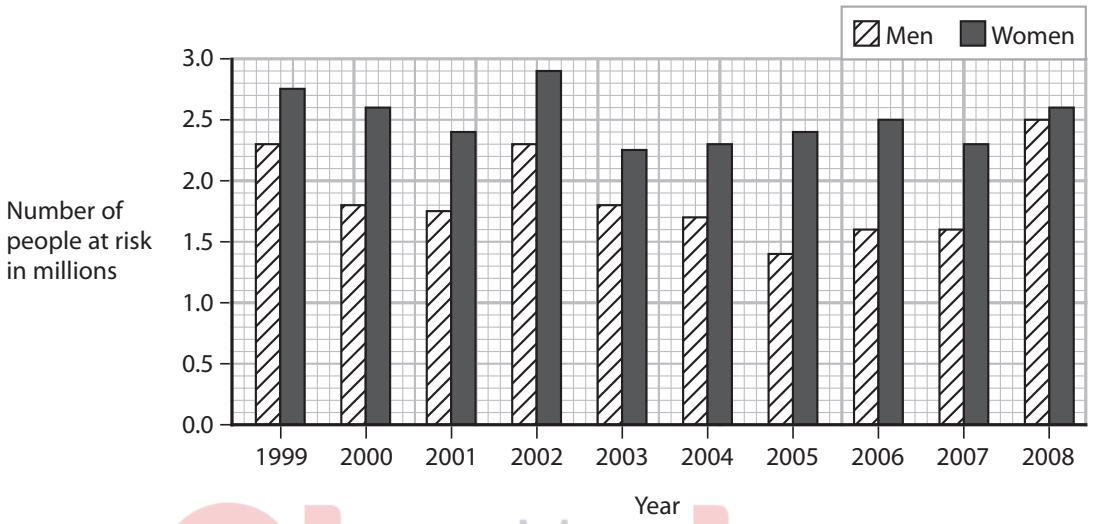
Working and answer

5. Question 2 · part a.iii · 2 marks

June 2022 Paper 2

Further practice · Attempt part a.iii only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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5. Question 2 · part a.iii · 2 marks

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Further practice · Attempt part a.iii only.

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(iv) Suggest one possible reason why the number of females at risk of developing a mental illness increased from 2007 to 2008.

(1)

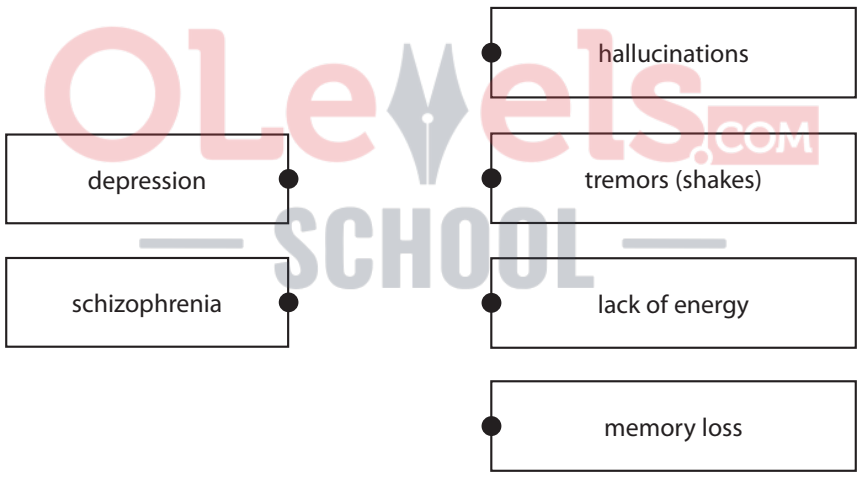
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(b) The boxes list two mental illnesses and some symptoms.

Draw one straight line from each mental illness to a symptom of the illness.

(2)



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(c) Which combination of drugs is most likely to affect mental health?

(1)

- ☒ **A** alcohol and heroin
- ☒ **B** antibiotics and alcohol
- ☒ **C** cannabis and paracetamol
- ☒ **D** paracetamol and antibiotics

(d) People who use the drug cocaine often have a raised body temperature.

Name two body structures that help to control body temperature.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

5. Question 2 · part a.iii · 2 marks · Working and answer

Working and answer

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6. Question 4 · part c · 2 marks

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Further practice · Attempt part c only.

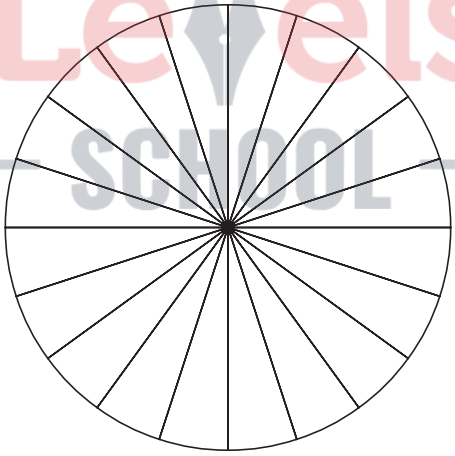
4 The table shows the percentage composition of the diets of two 10 year-old girls.

One girl lives in Area A and the other girl lives in Area B.

Component	Percentage (%) composition of diet	
	Area A	Area B
cereals	75.0	15.0
fruit and vegetables	15.0	37.5
eggs, fish and meat	2.5	27.5
milk, cheese and butter	7.5	20.0

(a) Complete the pie chart to show the diet of the girl in Area B.

(4)



(b) Explain what is meant by a balanced diet.

(2)

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6. Question 4 · part c · 2 marks

June 2023 Paper 1R

Further practice · Attempt part c only.

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(c) Calculate the ratio of the percentage composition of cereals in the diet of the girl in Area A compared with the diet of the girl in Area B.

Give your answer in the form n : 1

(2)

ratio = ...

(d) Discuss the effects that the percentage composition given for eggs, fish and meat in the diet may have on the growth of the two girls.

(5)

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(Total for Question 4 = 13 marks)

6. Question 4 · part c · 2 marks · Working and answer

Working and answer

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7. Question 3 · part a.i · 2 marks

November 2023 Paper 1

Further practice · Attempt part a.i only.

3 The table shows the flow of blood to different organs in a person at rest and during exercise.

	Blood flow in cm^3 per minute	
	at rest	during exercise
Heart	250	750
Skeletal muscles	1000	12 500
Skin	400	1900
Gut	1400	600
Brain	750	750
Total flow	5600	17 500

(a) (i) Give the ratio of the blood flow to the heart when at rest and during exercise.

Give your answer in the form 1 : n

(2)

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Ratio = ... : .....

(ii) Calculate the percentage increase in the total flow of blood between rest and exercise.

Give your answer to two significant figures.

(3)

percentage increase = ... %

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7. Question 3 · part a.i · 2 marks · Working and answer

Working and answer

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8. Question 4 · part c.i · 1 mark

November 2023 Paper 2

Further practice · Attempt part c.i only.

4 Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

Atorvastatin belongs to a group of medicines known as statins. Statins are medicines that regulate lipid levels in the blood. Statins are used when changes of diet and lifestyle have failed to lower blood cholesterol levels.

Statins can also be used to reduce the risk of heart disease when cholesterol levels are normal. People with liver disease or kidney problems should avoid taking statins. Women who are pregnant or are breastfeeding should also avoid taking statins.

In the United Kingdom approximately 7.5 million people take statins. This is 12% of the total population. However, up to 20% of these people stop taking statins because of adverse side effects such as muscle aches, fatigue, feeling sick and joint pains.

A recent study looked at patients who had stopped taking statins because of side effect symptoms. Patients were asked to take statins for four months, then placebo tablets (tablets that look like statins but contain no active substances) for four months followed by no tablets for four months. Every day the patients were asked how bad their symptoms were using a score from 0 to 100. The table shows the results.

Four-month treatment	Mean score out of 100
statins	16.3
placebo	15.4
no tablets	8.0

(a) Describe the changes in diet and lifestyle that could lower cholesterol. (Lines 2 and 3)

(2)

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8. Question 4 · part c.i · 1 mark

November 2023 Paper 2

Further practice · Attempt part c.i only.

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(b) (i) Explain why people with liver disease or kidney problems should avoid taking statins. (Lines 5 and 6)

(4)

(ii) Explain why women who are pregnant or breastfeeding should avoid taking statins. (Lines 6 and 7)

(3)

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(c) (i) Using the results of the study, calculate how many times greater the score for symptoms when taking statins is compared with the score when not taking tablets.

(1)

times greater = ...

8. Question 4 · part c.i · 1 mark

November 2023 Paper 2

Further practice · Attempt part c.i only.

(ii) Discuss the results of the study using information from the table.

(4)

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(Total for Question 4 = 14 marks)

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Working and answer

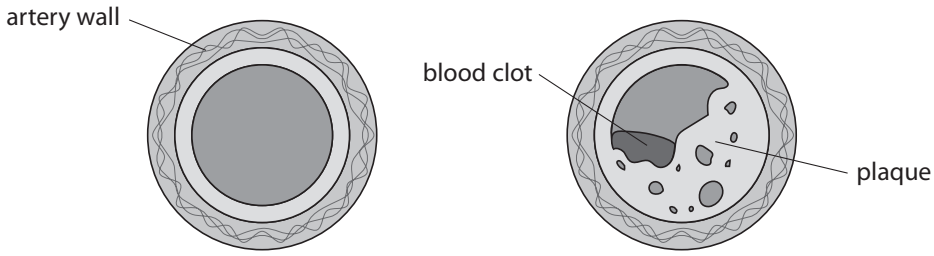
9. Question 6 · part b.ii · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

6 (a) Cholesterol is a fatty substance that can build up inside arteries forming waxy deposits called plaques.

The diagram shows a healthy artery and an artery with a plaque.



healthy artery artery with a plaque

(Source: adapted from https://www.dreamstime.com/artery-plaque-blood-clot-medical-illustration-diseased-compared-to-healthy-cross-section-diagram-shows-image120895536)

Explain how plaques might affect the heart of an individual.

(4)

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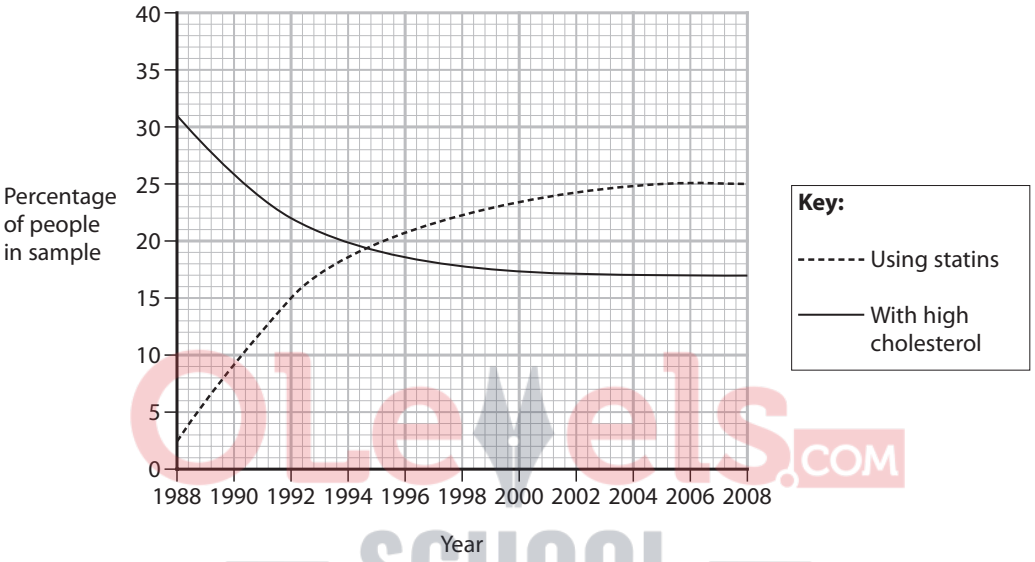
9. Question 6 · part b.ii · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

(b) Scientists investigated the effect of statins on cholesterol levels in a sample of people.

The graph shows the results of their investigation.



(Source: https://www.health.harvard.edu/heart-health/trends-in-high-cholesterol-and-statin-use)

(i) Give the relationship between the use of statins and the percentage of people with high cholesterol levels.

(1)

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(ii) Calculate the ratio of the percentage of people with high cholesterol levels in 1988 compared with the percentage of people with high cholesterol levels in 2005.

Give your answer in the form of 1 : n

(3)

ratio = ...

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9. Question 6 · part b.ii · 3 marks

June 2024 Paper 2R

Further practice · Attempt part b.ii only.

(iii) Describe a method that scientists could use to obtain the data shown in the graph.

(5)

(iv) Give two factors the scientists need to consider to make sure that their investigation is valid.

(2)

1 .....

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2 .....

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(Total for Question 6 = 15 marks)

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9. Question 6 · part b.ii · 3 marks · Working and answer

Working and answer

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10. Question 4 · part c.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.iii only.

4 Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

When looking at data, it is clear that people in lower income countries are more likely to die from infectious diseases (approximately 240 per 100 000) than non-infectious diseases (approximately 180 per 100 000). Infectious diseases such as diarrhoea and malaria can be easily eliminated or treated. It is also possible to vaccinate against diseases such as TB. These diseases are uncommon in higher income countries because of preventative medicine.

The only common diseases present in both higher and lower income countries are respiratory tract infections. These infections are usually influenza, pneumonia and bronchitis. Whilst influenza is a viral infection, the other infections are usually bacterial and can be treated. Many people at risk in the higher income countries are encouraged to get preventative treatment. These are often provided free of charge for the elderly who are most at risk and those who work in indoor spaces such as schools, offices and hospitals. The effect on mortality in lower income countries is more than two times as great as in higher income countries.

The greatest killers in higher income countries are heart disease and stroke. These two diseases kill approximately twice as many people per 100 000 as in lower income countries. These are often seen as lifestyle diseases. When we compare the number of deaths caused by non-infectious diseases, in higher income countries there are 445 deaths per 100 000 and in lower income countries there are 185 deaths per 100 000. There are also more cases of other diseases such as dementia and cancers in higher income countries.

(a) State one way the infectious diseases diarrhoea and malaria can be prevented (lines 3 and 4).

(2)

diarrhoea

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malaria

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10. Question 4 · part c.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.iii only.

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(b) (i) State how pneumonia and bronchitis can be treated (lines 8 to 10).

(1)

(ii) Give one reason why elderly people and people who work in indoor spaces are at risk from respiratory tract infections. (lines 12 and 13).

(2)

elderly people

indoor spaces

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(c) (i) Give two factors that may contribute to lifestyle diseases (line 18).

(2)

1 .....

2 .....

(ii) In a higher income country of 5 000 000 people, calculate the number of people who will have non-infectious diseases (lines 20 and 21).

Give your answer to three significant figures.

(3)

number of people = ...

10. Question 4 · part c.iii · 3 marks

November 2024 Paper 2

Further practice · Attempt part c.iii only.

(iii) Calculate the ratio of the number of people with non-infectious diseases in a higher income country to the number of people with non-infectious diseases in a lower income country (lines 20 and 21).

Give your answer in the form n : 1

(3)

ratio = ... : .....

(iv) Suggest why there are more cases of dementia and cancers in higher income countries (lines 21 and 22).

(2)

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(Total for Question 4 = 15 marks)

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10. Question 4 · part c.iii · 3 marks · Working and answer

Working and answer

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11. Question 4 · part a.ii · 2 marks

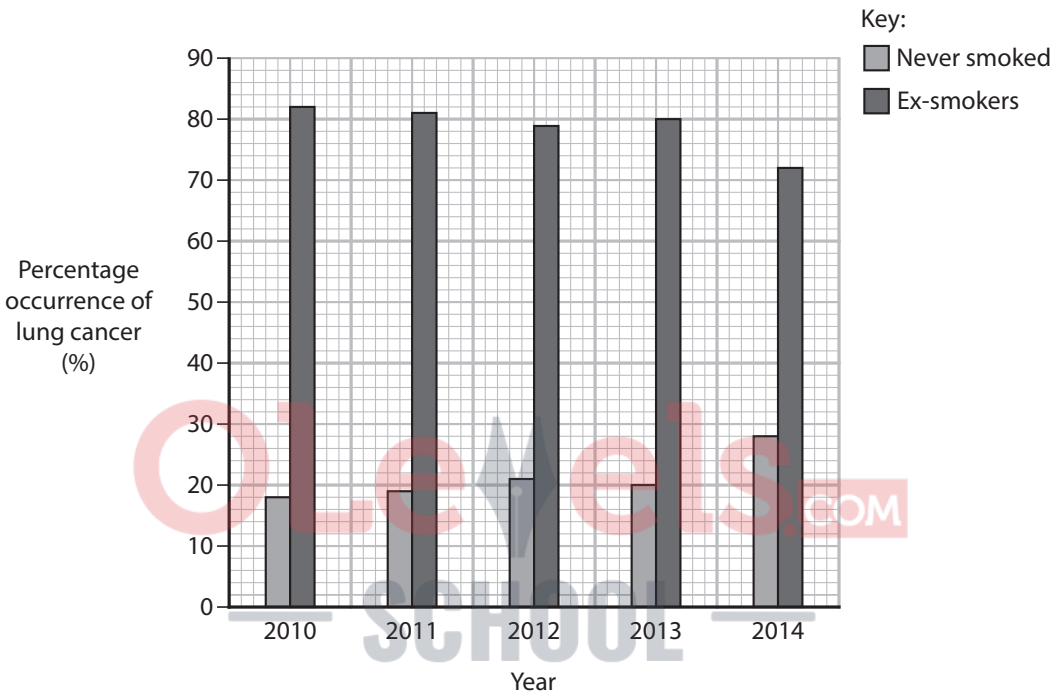
June 2025 Paper 2R

Further practice · Attempt part a.ii only.

4 (a) Cigarette smoking is the biggest cause of lung cancer worldwide.

The bar chart shows the percentage occurrence of lung cancer, from 2010 to 2014, in ex-smokers and in people who have never smoked cigarettes.

People who still smoke are not included in the bar chart.



(Source adapted from: https://www.thetimes.co.uk/article/pollution-blamed-for-lung-cancer-in-people-who-have-never-smoked-r52fr7m89)

(i) Describe the trend in the percentage occurrence of lung cancer in ex-smokers, from 2010 to 2014.

(2)

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11. Question 4 · part a.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part a.ii only.

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(ii) Calculate the ratio of the percentage occurrence of lung cancer in ex-smokers to people who have never smoked, in 2012.

Give your answer in the form n : 1

(2)

ratio = ... : 1

(iii) Suggest two reasons for the difference in the percentage occurrence of lung cancer, from 2010 to 2014, in people who have never smoked, compared with ex-smokers.

(2)

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(iv) Name the substance in cigarette smoke that causes lung cancer.

(1)

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11. Question 4 · part a.ii · 2 marks · Working and answer

Working and answer

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